

Advanced (Carolina Reaper)

- Q1.** Graph the linear inequality $2x + 3y \geq 12$.
 (A) Solid line with shading above
 (B) Dashed line with shading below
 (C) Solid line with shading below
 (D) Dashed line with shading above
 (E) No shading
- Q2.** The boundary line of the inequality $x - 2y < 4$ is
 (A) Dashed
 (B) Solid
 (C) Vertical
 (D) Horizontal
 (E) Diagonal
- Q3.** The solution region of the inequality $y \geq -2x + 1$ is
 (A) Below the line
 (B) On the line and above
 (C) On the line and below
 (D) Above the line
 (E) Nowhere
- Q4.** Tom has 120 to spend on video games and snacks. Video games cost \$20 each and snacks cost \$5 each. The inequality $20v + 5s \leq 120$ represents
 (A) The cost of video games
 (B) The cost of snacks
 (C) The total cost of video games and snacks
 (D) The number of video games and snacks
 (E) The time spent playing video games
- Q5.** The inequality $3x + 2y \leq 12$ can be rewritten as
 (A) $y \leq -\frac{3}{2}x + 6$
 (B) $y \geq -\frac{3}{2}x + 6$
 (C) $y \leq \frac{3}{2}x + 6$
 (D) $y \geq \frac{3}{2}x + 6$
 (E) $y = -\frac{3}{2}x + 6$
- Q6.** The graph of the inequality $x + 2y > 4$ has a boundary line with a slope of
 (A) $-\frac{1}{2}$
 (A) $\frac{1}{2}$
 (A) -2
 (B) 2
 (C) 1
- Q7.** The inequality $2x - 5y < 10$ can be graphed by first graphing the equation
 (A) $2x - 5y = 10$
 (B) $2x + 5y = 10$
 (C) $2x - 5y = -10$
 (D) $2x + 5y = -10$
 (E) $x - 2y = 5$
- Q8.** A bookshelf has 5 shelves, and each shelf can hold at most 8 books. The inequality $5s \leq 40$ represents
 (A) The number of books on each shelf
 (B) The total number of books on all shelves
 (C) The number of shelves
 (D) The cost of the books
 (E) The weight of the books

Open Ended Questions (FRQ)

- Q9.** Graph the system of linear inequalities: $x + 2y \geq 4$ and $3x - 2y \leq 6$. Shade the solution region.
- Q10.** A bakery sells a total of 250 loaves of bread per day. The inequality $3b + 2c \leq 250$ represents the number of whole wheat bread loaves (b) and white bread loaves (c) sold. Graph the inequality and find the solution region.

Chef's Tips

- Tip 1: Ensure students understand the difference between solid and dashed boundary lines.
- Tip 2: Remind students to shade the solution region carefully.
- Tip 3: Encourage students to check their work by plugging in test points.